MyDesign Portfolio
Element E Initial Template

STEM Principle for High-Priority Design Requirement #1

STEM Concept: **The shelter will be 10 x 20 inches**
Describe how STEM concept applies directly to the specifics of your design:
Our group will use math to create a shelter that fits the criteria of being 10 x 20 inches

STEM Principle for High-Priority Design Requirement #2

STEM Concept: **The shelter must support a weight up to 10 pounds**
Describe how STEM concept applies directly to the specifics of your design:
Our group will use our engineering skills to create a structure that is cost efficient, yet remains strong

STEM Principle for High-Priority Design Requirement #3

STEM Concept: **There must be a roof in order to regulate sunlight**
Describe how STEM concept applies directly to the specifics of your design:
Our group will use our science backgrounds to understand how sunlight affects our plant

STEM Principle for High-Priority Design Requirement #4

STEM Concept: **The shelter will be battery operated and use solar charging**
Describe how STEM concept applies directly to the specifics of your design:
Our group will use technology to rig our shelter to be battery operated

STEM Principle for High-Priority Design Requirement #5

STEM Concept: **The shelter must be programmable**
Describe how STEM concept applies directly to the specifics of your solution:
Our group will use technology to use coding platforms to control the necessary components needed to control our shelter

STEM Principle for Design Solution #1

STEM Concept: **Functionality**

Describe how STEM concept applies directly to the specifics of your solution:
STEM concepts apply to this because the shelter needs to be functional and in order to do so our group must use science, technology, engineering, and math

STEM Principle for Design Solution #2

STEM Concept: **Efficiency**

Describe how STEM concept applies directly to the specifics of your solution:
STEM directly applies to this because our team needs to use engineering in order to maximize space and use minimal materials to keep costs low

STEM Principle for Design Solution #3

STEM Concept: **Durability**

Describe how STEM concept applies directly to the specifics of your solution:
Our teams shelter needs to be strong and long lasting, so we use science to understand what materials work best for us, and we use engineering to create a sturdy structure